2407011 16

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NAME: G. Dhanushiya STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: _____

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Ex-1

Basic Linux Commands

Aim:

To study various network commands used in Linux & Windows

Commands

1] arp -a :- It will show IP address of your computer along with IP address of your router

output :- gatway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3

2] Hostname :- It displays name of computer

output :- fedora

3] ipconfig /all :- It displays detailed configuration information of your TCP/IP connection.

output :-

Hostname → HDC0422267

Physical address → 88-AE-DD-15-ED-46

IPV4 address → 172.16.10.192

DHCP enabled → no

4] nbtstat -a :- It helps solve problems with NetBIOS

output :- NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

5] netstat -r :- It displays variety of statistics about computer TCP/IP connections.

output:-

network destination	netmask	gateway	interface	metric
0.0.0.0	0.0.00	172.16.81	172.16.10.192	281

6] nslookup :- It is used to display DNS details

output :-

Server : unknown
Address : 172.16.81
non-authoriative answer :
name : www.google.com
Address : 2001:4860:482c:7700

7] pathping :- Combination of ping and tracert commands.

output :- Ussage : pathing [-g host-list] [-h maximum-hops] [-i address] [-n] [-p period] [-q num-queries] [-4] [-6] target_name

8] ping :- It is best way to test connectivity blw nodes

* ping localhost

pinging HDC04222671 [::1] with 32 bytes of data
Reply from ::1 : time <1ms

* ping 4.2.2.2

Reply 4.2.2.2 bytes=32 time=210ms TTL=55

* ping facebook

Reply from 57.144.51.1 : byte=32 times=4ms TTL=55

Some Important Linux networking commands

1. ip

The ip command can show address information, manipulate routing, plus display network various device.

ip < options > < object > < command >

a) ip address show : To show IP address

output :-

1: lo: < LOOPBACK, UP, LOWER_UP > mtu 65536 q disc noqueue state UNKNOWN group default q len 1000

link / loopback 00:00:00:00

inet 127.0.0.1 / 8 scope

b) ip address add 192.168.1.254 / 24 dev enp503

output:-

enp053 : < BROADCAST, MUTICAST, UP, LOWER_UP > mtu 1500

link / ether 08:00:27:18:53 brb ff:ff:ff:ff

inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic

inet 192.168.1.254/24 scope global enpos3

c) To delete an IP address :- ip address del 192.168.1.254/24 dev enps03

output:-

enpos3 : < BROADCAST, MUTLI CAST, UP, LOWER > mtu 1500

link / ether 08:00:27:18:53 brd ff:ff:ff:ff

inet 10.0.2.15 / 24 brd 10.0.2.255 Scope global dynamic

d) Alter the status of interface :- ip link set eth0 up

output:-

enp0s3: <BROADCAST, MUTLICAST, UP, LOWER> mtu 1500 qdisc fq-codel state UP mode

link/ether 08:00:27:87:18:53 brd ff:ff:ff

altname enx080027871853

e) Alter the status of interface :- ip link set eth0 down

output:-

enp0s3 : <BROADCAST, MUTICAST, UP, LOWER> mtu 1500 qdisc fq-codel state down mode

link/ether 08:00:27:87:18:53

altname enx080027817853

f) Alter status by enabling promiscuous mode

ip link set eth0 promisc on

output:-

enp0s3 : <BROADCAST, MULTICAST, PROMISC> mtu 1500 qdisc fq-codel DOWN mode default group

link/ether 08:00:27:87

altname enx080027871853

g) Add a default route.

ip route add defaut via 192.168.1.254 dev eth0.

output:-

default via 192.168.1.254 dev eth0

192.168.1.0/24 dev eth0 kernel scope link src 192.168.1.10.

2] ifconfig :- Used for configuring and troubleshooting network

output :-

lo : flags = 73 <UP, LOOPBACK, RUN > mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10 <host>

3] mtr :- It is a program with command-line interface that serves as a network diagonsis tool.

output :- mtr google.com

Host : server		Lost	Snt	Last	Avg
1.	192.168.1.1	0.0%	10	1.2	1.4

mtr -c 10 → send 10 pings

Host : server		Lost %	Snt	Last	Avg
1.	192.168.1.1	0.0%	10	1.1	1.3

Ex-2. Study of network cables

Aim: To study different types of network cables

a) Understand different types of cable

Cable Type	Category	Maximum data Transmission	Advantage Disadvantage	Use
UTP	Category 3	10 bps	Advantage → cheaper → easy to install	10 Base-T ethernet
	Category 5	up to 100 mbps	Disadvantage → more prone to (EMI) electromagnetic noise	Fast ethernet gigabit ethernet
	Category 5e	1Gbps		Fast ethernet, gigabit ethernet
STP	Category 6, 6a	10 Gbps	Advantage → shielded → Faster → Less susceptible to noise	Gigabit ethernet 10G ethernet, (55m) widely used in data centre
SSTP	Category 7	10GB	Diadvantage → Expensive → greater instalation effort	Gigabit ethernet 10G ethernet (100m)
			→ High bandwidth → Low Loss bandwidth → Versatile	Speed of signal is 300m TV network

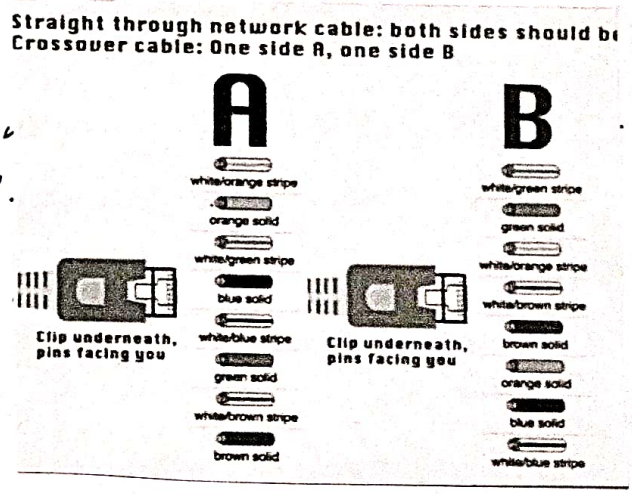
Coaxial cabel	RG-6 RG-59 RG-11	10-100 Mbps	Disadvantage → Limited distance → Cost → Size is bulky
fibre optics cable	single mode multi mode	100Gb PS	Adv → high speed → high security disadvantage → Expensive

Maximum distance of fibre optics cable around 100m.

b) Make your own ethernet cross-over cable

Tools and parts needed:-

* ethernet cabling. CAT5E is .
* Crimping tool. This is an all. shapped to push down the
* Two RJ45 plugs
* optional 2 plug shields.



Step 1: To Start construction of the device, begin by threading shields onto the cable.

Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

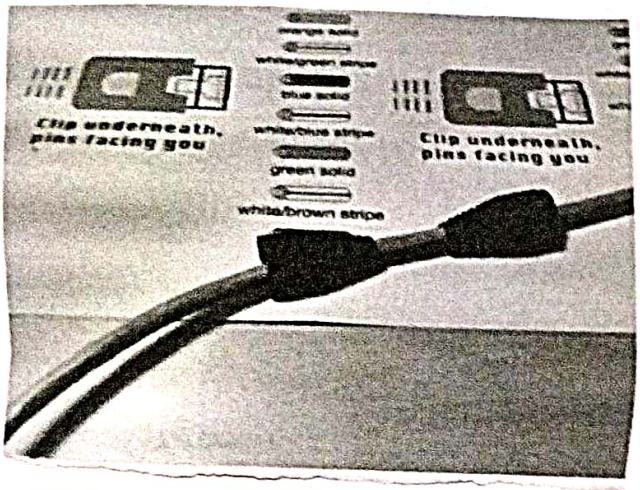
tep 3: After, you will need to untangle the wires; re should be four 'twisted pairs'. Refer encing k to sheet, arrange them from top to bottom.

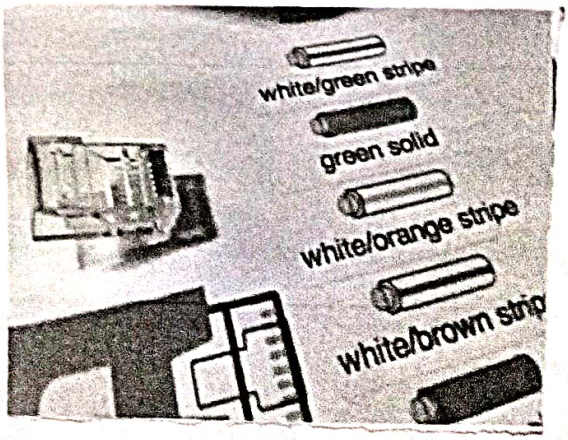
Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others. Hold the plug with the clip side facing away from you.

Step 5: Next, push the cable right in. The notch at the end of the plug needs to be over the cable.

Step 6: After the wires are securely sitting inside the plug, insert it into tool and push down

Step 7: Last, repeat for other end using diagram B.





Result: Thus, the experiment is executed successfully.

Ex-3 Experiment on CISCO PACKET TRACER

Aim:-
To study the packet tracer tool installation and user interface overview

c) To understand envisonment of CISCO PACKET TRACER to design simple network.

Introduction:
A simulator, as name suggests, stimulates network devices and its envisonment.

1. It allows you to model complex systems without the need for dedicated equipment.
2. It helps you to practice your network configuration.
3. It is available for Linux and windows desktop.
4. Protocols in Packet tracer are cooled to work and behave in same way.

Installing Packet Tracer:
To download it, go to https://www.netacad.com and login in with your Cisco Networking Credentials.

Windows:-
Installation in windows is pretty simple and straightforward; the setup comes in single file. Open the file to begin the setup wizard, accept the license agreement, choose a location and start the installazation.

Linux :-

Linux users with an Ubuntu / Debian distribution should download the file for Ubuntu, and those using Fedora must download. Grant executable permission to this file by using chmod.

User Interface overview :-

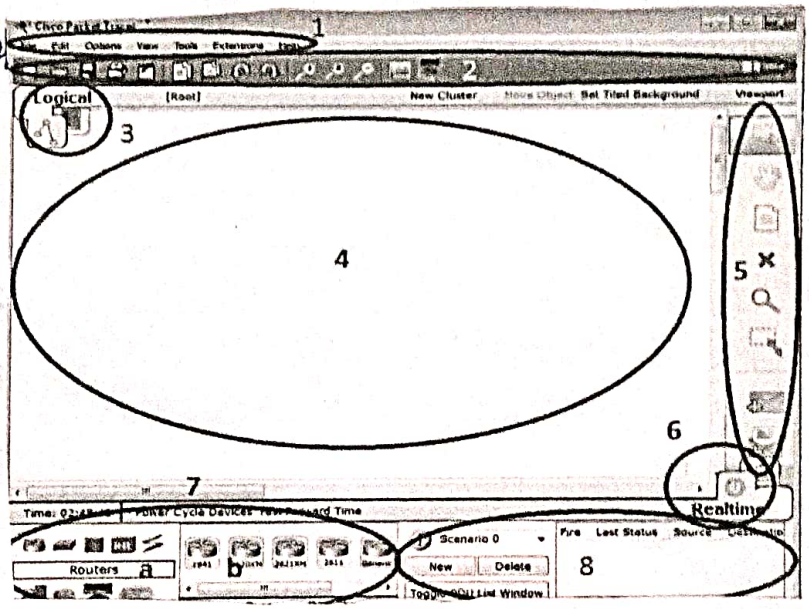
1. Menu Bar - This is a common menu found in all software applications. It is used to open, save, print, change prefences and so on.
2. Main Toolbar - This bar provides shortcut icons to menu options that are commonly accessed such as open, save, undo.
3. Logical, physical workspace tab - These tabs allow you to toggle between logical and physical work areas.
4. Workspace - This is area where topologies are created and simulation are displayed.
5. Common tools bar - This toolbar provides control for manipulating topologies such as select, move, place note, delete, inspect, resize shape and add simple, complex PDU.

6. Real-time / simulation tabs - These tabs are used to toggle b/w real and simulation tabs modes.

7. Network component box - This component contains all of the network and end devices available with packet Tracer.

8. Searched packet box - Users can create highly customized packets to test their topology

from th



d) Analyse the behaviour of network devices using CISCO PACKET TRACER simulator.

1. From network component box, click & drag and drop the below components.

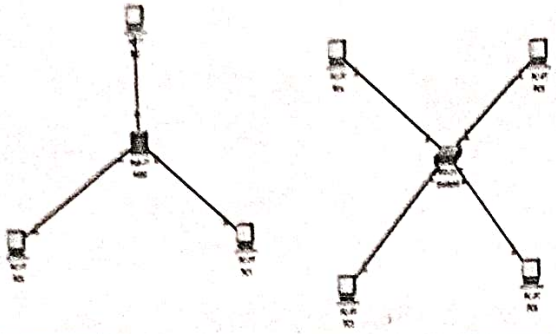
   a. 4 Generic PC's and 1 HUB
   b. 4 Generic PC's and 1 switch.

2. click on connections:-

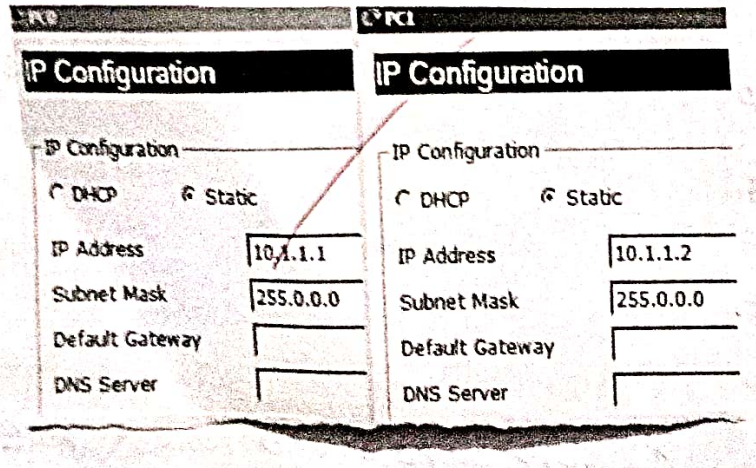
a. click on copper straight - Through cable.

b. select one of the PC and connect it to HUB using the cable. The link LED should glow in green, indicating that the link is up.

c. Similarly connect HUB to switch.



3. click on desktop tab, click on IP configuration and enter an IP address and subnet mask.



click on PDU from common tool bar.

a) Drag and drop it one of Pc.

4. Observe the flow of PDU from source PC to destination PC by selecting the real time mode of simulation.

5. Repeat step # 3 to step # 5 for the PC's connected to switch.

6. observe how HUB and switch are forwarding the PDU and write your observation.

Result:- Thus experiment for CISCO platter network is done using simulation tool.

Ex-4

Local Area Network.

Aim:

To setup and configure LAN using a switch and ethernet cables in you lab.

What is a LAN?

Step 1:- Plan and design an appropriate network topology taking into account network requirements and equipment location.

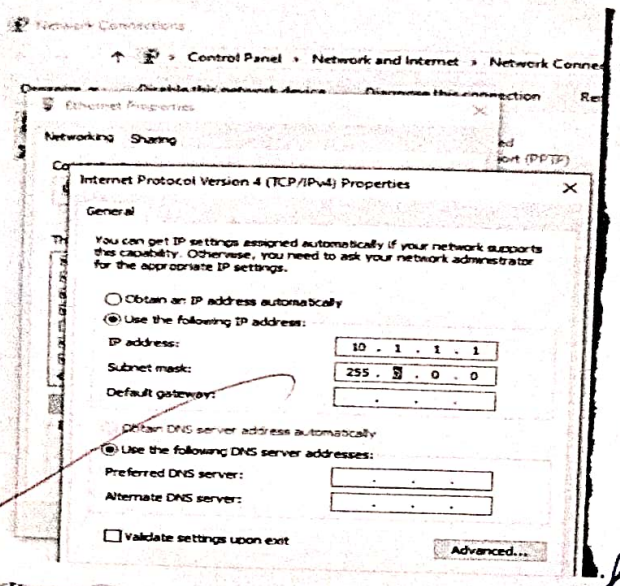
Step 2:- You can take 4 computer - a switch with 8, 12, 24 ports which is sufficient for network of these sizes and 4 ethernet cables.

Step 3:- Connect your computers to network switch via an ethernet cable, which is as simple as plugging one and of the ethernet cable into your computer.

Step 4:- Assign IP address to your PC's

a) Log on to the client computer as owner
b) click network and internet connections.
c) Right click LAN / ethernet → go to prop → select internet protocol → click on

properties → select use the following IP address option and assign IP address.



Step 5:- Configure a network [illegible] :

1. Connect your computer to the switch. To access the switch's web interface, you'll need to connect your computer to the switch using an ethernet cable.

2. Assign IP address as :- 10.1.1.5
   subnet mask 255.0.0.0

Step 6:- check the connectivity blw switch and other machine by using ping command is the command promt on the device.

Step 7:- Select a folder → go to prop → click sharing tab → share it with everyone on the same LAN.

Step8:- Try to access the shared folder from others computers of the network.

Result:

Thus the setup of Local area network using a switch and ethernet is executed.

Ex-5

Capture Tool : Wireshark.

Aim:- Experiments on Packet Capture tool : Wireshark

Packet sniffer :-

* Sniffs message being sent / received from computer
* Stores and display contents of various protocol fields in the messages.
* Passive program
  - → never sends packets itself
  - → no packet addressed to it
  - → receives a copy of all packets

Packet sniffer Structure :-

* Tcpdump
  - eg:- tcpdump -enx host 10.19.24
* Wireshark
  - eg:- wireshark -r

Description:-

Wireshark, a network tool, captures packets in real time and display them in human-readable format. It includes filters, color coding and let you dig deep into individual packets

What we can do with it :-

- → capture network traffic
- → Decode packet protocol using dissectors
- → define filters - capture & display

→ watch smart statistics

Wireshark used for:

→ Network administrations : troubleshoot

→ Network security engineers : examine problem

→ developers : debug

→ people : learn protocol

Getting wireshark :

Wireshark can be downloaded for windows from website. For Linux, it will be found in it package repositories. For ubuntu, it is found at ubuntu centre

Capturing packets :

After downloading, launch it and double click the name of network under capture to start capturing packets on it.

Packet List Pane :-

The packet list pane displays all packets in the file. This each line in packet list to one packet in capture file. If you select a line, more details will be displayed.

Packet details pane :-

The packet details pane shows the current packet in a more detailed form. This shows the protocol files of the packet selected in a "packet list" pane. These protocol and fields can be expanded.

Packet bytes pane :-

The packet bytes pane shows the data of current packet in a hexdump style.

Color coding :-

Wireshark uses colors to help you identify the types of traffic at a glance. By default, light purple is TCP traffic, light blue is UDP and blacks is for packets with errors.

To view exactly what color codes mean, click view > coloring rules. You can also customize and modify coloring rules from here.

Sample captures :-

If there's nothing interesting on your network, wire-shark's wifi has you covered. The wiki contains a page of sample captures files. Click file > open in wireshark and open the downloaded file.

Filtering packets :-

If you are trying to inspect something specific, it helps to close down all other applications using network so you can narrow down traffic.

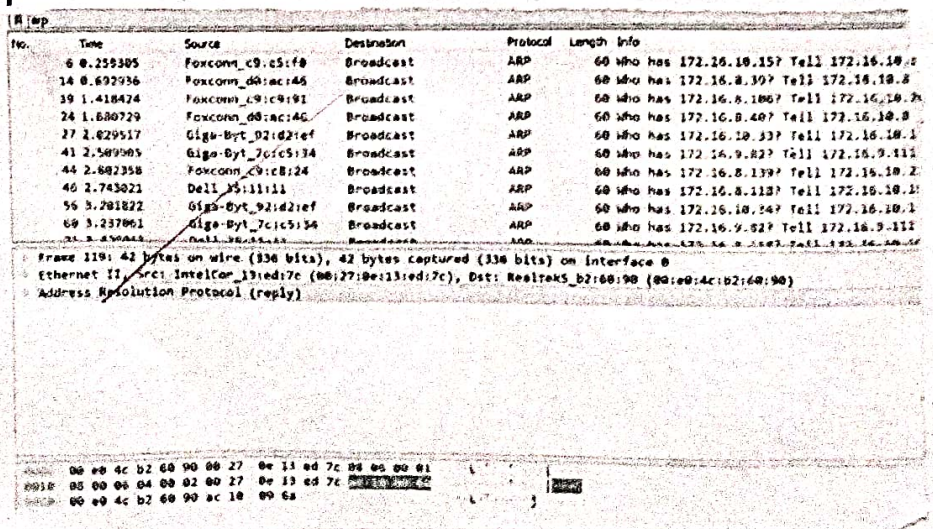
Capturing and Analysing packets using Wireshark tool.

To filter, capture, view, packets in tool. Capture 100 packets from ethernet : IEEE 802.3 LAN interface

Procedure:-

→ Select Local area connection in it
→ Go to capture option
→ Select stop capture after 100 packets
→ Then click start capture
→ Save the packets.

O/p

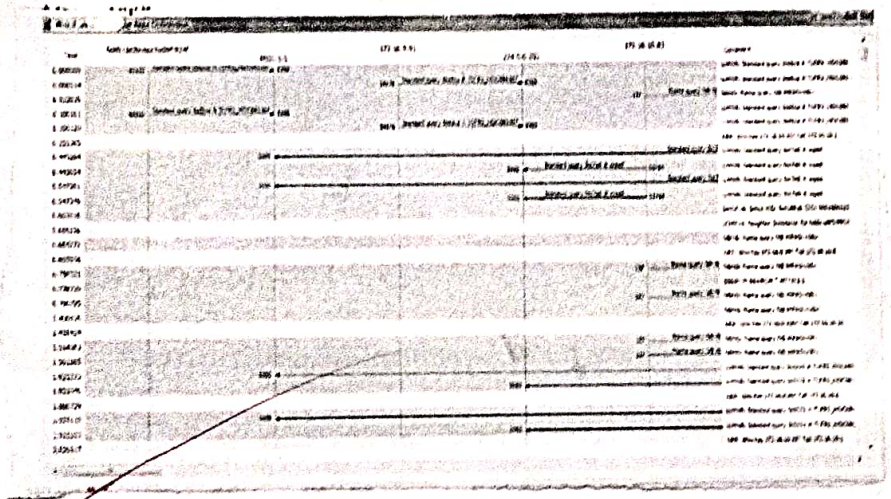


1. Create a filter to display only TCP / UDP packets

Procedure:

→ Select Local area connection
→ Go to capture option
→ Select stop capture after 100 pkts
→ Then start capture
→ Search TCP packets
→ To see flow graph click stastistic flow graph
→ Save packet

O/P:-

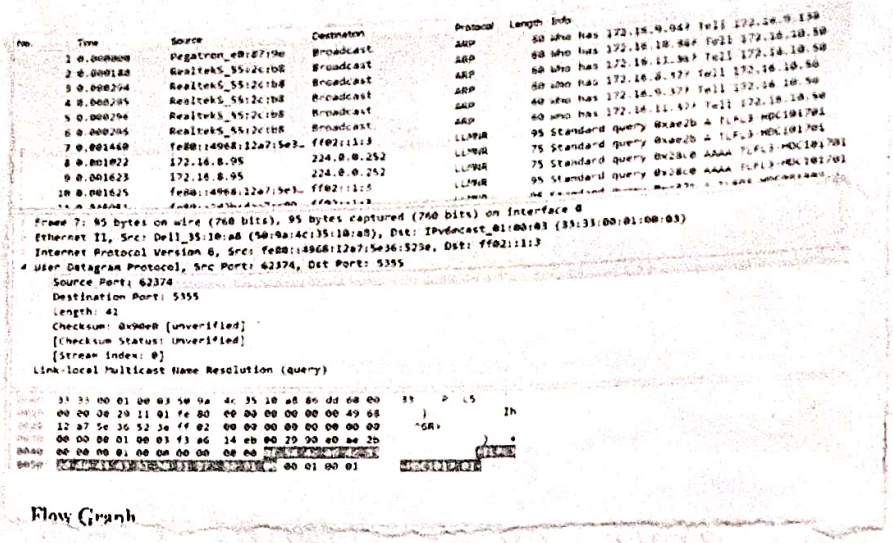


2. Create filter to display only ARP packets :-

Procedure:

→ Go to capture option
→ Select stop capture after 100 pkts
→ Click start capture
→ Search ARP packets
→ Save the packets.

O/P:-

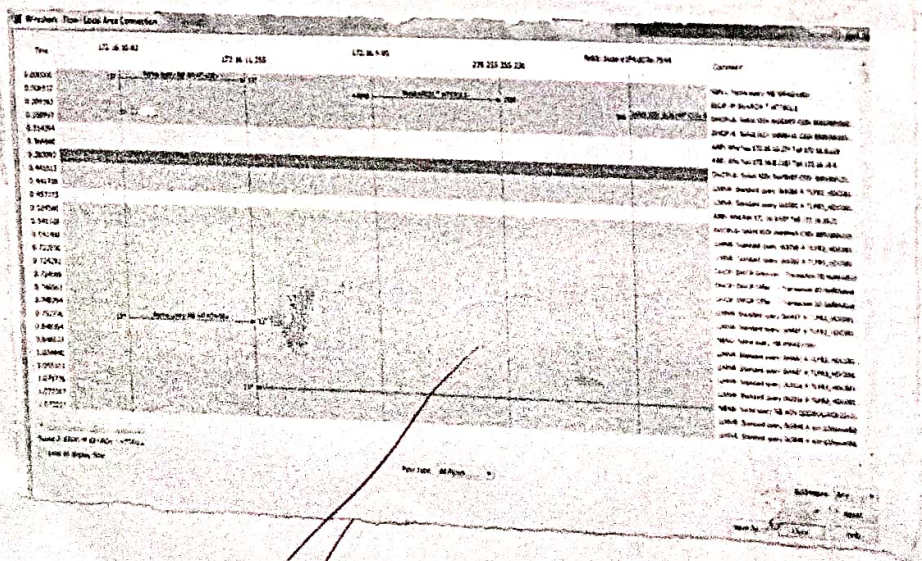


3. Create a filter to display only DNS

Procedure :

→ Go to capture option
→ Select stop capture after 100 pkts
→ click start
→ Search DNS capture and flow graph.
→ Save packets.

O/P:-

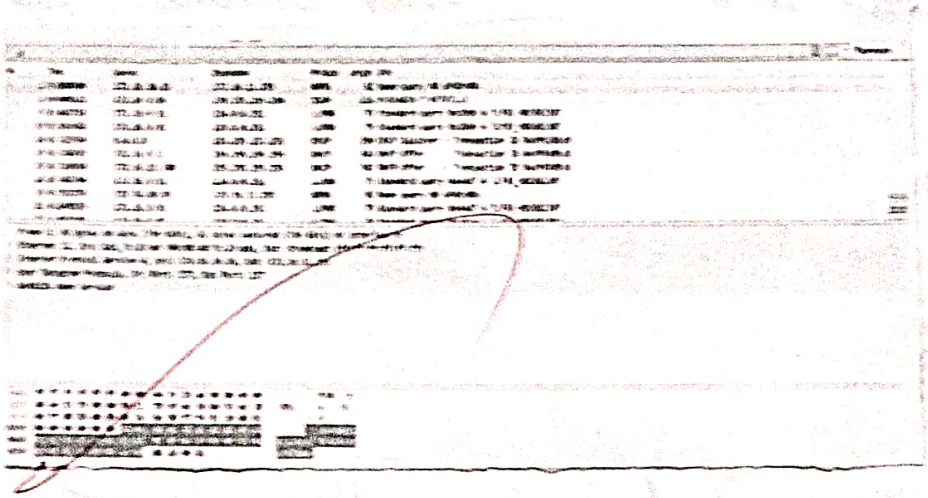


4. Create filter to display only HTTP packet

Procedure:-

→ Selet local area network
→ Go to capture option
→ Selet stop capture & click start capture
→ Search HTTP packets and save it

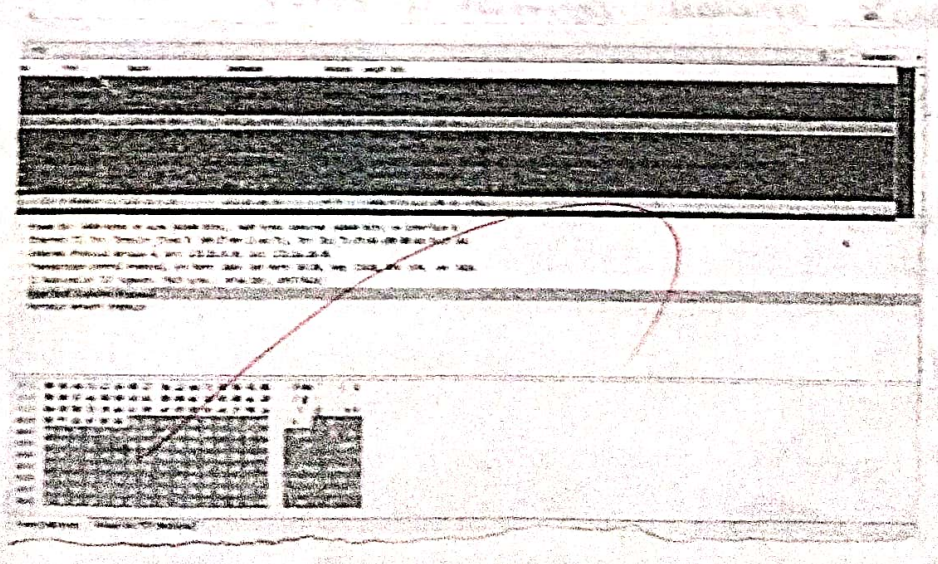
O/P:-



5. Create filter to display only IP/ICMP

→ Select local area network
→ Go to capture option
→ Search ICMP/IP packet in search bar.

o/p:-



Result:- Thus an experiment on capture tool using wireshark is done successfully.

Ex-6

Error Correction in Data Link Layer

Aim: To write a program to implement error detection and correction using Hamming code concept.

Error correction at data link layer :-

Hamming code is a set of error correction code that is used to detect & correct errors.

Create sender program :-

1] Input to sender file should be text. Program convert text to binary
2] Apply Hamming code on binary data and add bits
3] Save this output in file called channel.

Create receiver program :-

1] Program read the input from channel file
2] Apply hamming code on data to check error
3] If error, display position of error
4] Else remove bits and convert to ascii.

Code:- Sender

```
data = input ("enter 4 bits:")
d1, d2, d3, d4 = map (int, data)
p1 = d1 ^ d2 ^ d4
p2 = d1 ^ d3 ^ d4
p4 = d2 ^ d3 ^ d4
code = [p1, p2, d1, p4, d2, d3, d4]
```

```
Print ("Hamming code :". ".join (map (str, code )))
```

Receiver:

```
r = list (map (int, input ("enter received 7 bit:" )))
r1 = r[0] ^ r[2] ^ r[4] ^ r[6]
r2 = r[1] ^ r[2] ^ r[5] ^ r[6]
r3 = r[3] ^ r[4] ^ r[5] ^ r[6]
error = r1 * 1 + r2 * 2 + r4 * 4
if error == 0 :
    print ("no error")
else:
    print ("error at position:", error)
    r[error - 1] ^ = 1
    print ("Correct code :", ".join (map (str, received )))
```

Input:

```
Enter 4 data bit : 1011
Hamming code : 0110011
Enter received 7 bits : 0010011
```

Output:

```
Error at position : 2
Correct code : 0110011
```

Result: Error correction using Hamming code was executed successfully.

Ex-7

Flow control at data link layer

Aim:

To write a program to implement flow control at data link layer using sliding window protocol. Simulate the flow of frames from one node to other

Procedure:

Create a sender program:-

1. Input window size from user.
2. Input a text message from user.
3. Consider 1 character per frame.
4. Create frame with fields
5. Send the frames.
6. Wait for acknowledgment from receiver.
7. Reader a file called receiver_buffer
8. Check Ack field for acknowledgement no.
9. If number is as expected, send new sets of frames, else if NACK is received, resend frames

Create a receiver file:-

1. Reader a file called sender_buffer
2. Check frame no.
3. If frame no. are as expected, write ack no. in receiver_buffer file, else write NACK no.

Code:

```
import time
w = int(input("enter window size:"))
m = input("enter msg : ")
start = 0
while start < len(m):
    print("Sender")
    end = start + w
    if end > len(m):
        end = len(m)
    sender_b = []
    for i in range(start,end):
        f = [i, m[i]]
        sender_b.append(f)
        print("Frame no:", i, "Data:", m[i])
    print(sender_b)
    print("waiting for receiver..")
    time.sleep(1)
    print("receiver")
    ex = start
    er = False
    for f in sender_b:
        f_no = f[0]
        d = f[1]
    print("received frame:", f_no, "Data:", d)
        if f_no == ex:
            ex = ex + 1
        else:
            er = True
            break
```

E

```
        if er :
            received_buffer = ('NACK', ex)
        print ("Completed")
```

Input:

Enter window size : 4
Enter msg : HELL

O/P:-

Sender

Frame no : 0 Data : H
Frame no : 1 Data : E
Frame no : 2 Data : L
Frame no : 3 Data : L

Sender buffer : [(0,'H'), (1,'E'), (2,'L'), (3,'L')]

Waiting for receiver...

Receiver

received Frame no : 0 Data : H
Frame no : 1 Data : E
Frame no : 2 Data : L
Frame no : 3 Data : L

Received_buffer : ('ACK', 3)

Completed.

Result:

Thus, flow control using sliding window protocol was executed successfully.

x-8

NMAP to discover live hosts using NMAP Scans

Aim:

To discover live hosts on a network using Nmap host-discovery scans, ARP scans, ICMP scan & TCP/UDP ping scan

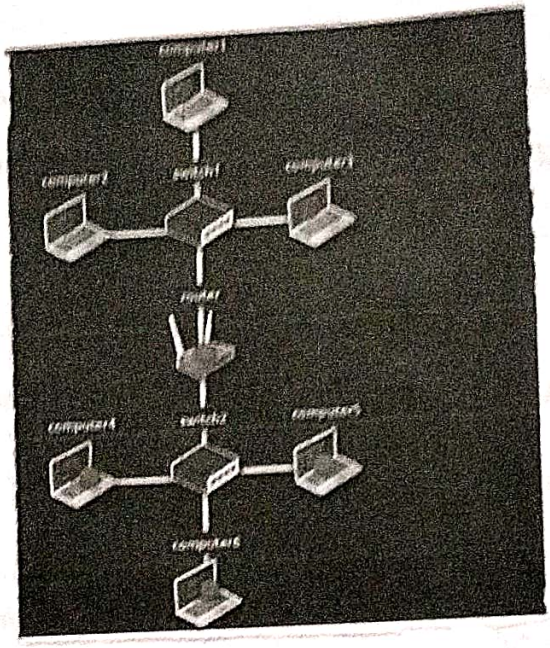
Nmap:

It is a free open-source network scanning tool created by Garden Lyon and released in 1997

Procedure:

1. Log into Tryhackme and open Nmap live host discovery
2. Work through task 1-3 and answer simulator-based questions
3. Deploy lab networks in Task 4 and start Attackbox; note target subnet.
4. Run ARP scan: sudo nmap -PR -sn <subnet>/24
5. Run as ICMP echo scan: sudo nmap -PE -sn <subnet>/24
6. Run ICMP timestamp (-PP) and Address-mask (-PM) scan
7. Run TCP SYN ping (-PS), a TCP ACK ping and UDP ping (-PU); compare results.
8. Optionally capture traffic with tcpdump or wireshark on tun0 interface.
9. Try nmap -sL <subnet>/24 and nmap -R <subnet>/24 to see target and reverse DNS.

10. Complete room's knowledge - check questions in Task 6-9, then move on to "Nmap Basic port Scan" room.



Result:

Thus Nmap host discovery & scan. were executed successfully using TCP, SYN, ACK, UPD ping.

Ex-7

Implementation of subnetting in cisco packet tracker stimulator

Aim:

To implement the concept of subnetting in cisco packet tracker stimulator by creating multiple subnets, assign IP addresses using router.

Description:

Subnetting is the process of dividing one network into smaller network called subnet. It helps in efficient use of IP address and allow multiple network to be created from a single network address.

Requirements:

* cisco packet tracer
* router
* switches
* PCs
* Copper straight through cables.

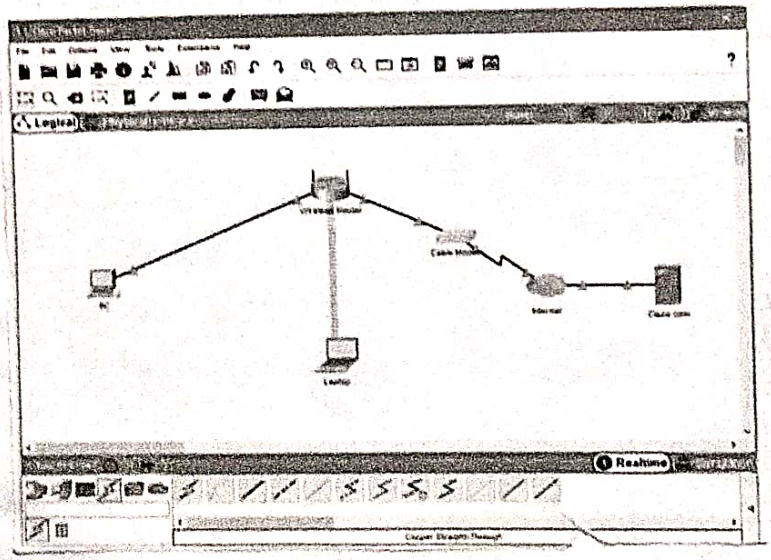
Procedure:

1. Open cisco packet tracer and create new network topology.
2. Add routers, switches and PCs.
3. Connect PCs to switch to router using cables.
4. Divide network 192.168.1.0/24 into smaller subnet using 127 subnet mask.

5. Assign static IP address to all PC.
6. Configure router interface through the CLI.

eg: enable
configure terminal
IP address 192.168.1.1
no shutdown
exit

7. Configure IP address, subnet mask and gateway on each PC.
8. Check connection btw PCs using ping command.
9. Send simple PDU btw PCs in different subnet
10. Observe packet transmission with & without router.



Result:

Thus, subnetting was successfully implemented in cisco packet tracer.

Ex-10

Internetworking with routers in cisco packet tracer

Aim:

To design and configure simple internetwork using router and communication btw 2 networks.

Requirement:

- ☆ cisco packet tracer
- ☆ Router
- ☆ 2 PCs
- ☆ copper cables.

Description:

A router is a networking device used to connect diff network and forward packet. If 2 PCs are connected to 2 interface of router, static IP address are assigned to PCs.

IP configuration

Device	IP address	Subnet mask	Gateway
Router Fa0/0	192.168.10.1	255.255.255.0	–
Router Fa0/1	192.168.20.1	255.255.255.0	–
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Procedure:

1. Open cisco packet tracer.
2. Place 1 router and 2 PCs.
3. Connect PC0 to router 1 FA 0/0 using cable.
4. Connect PC1 to router 1 FA 0/1 using cable.
5. Open router CLI and enter
6. Configure PC0:

   IP: 192.168.10.2

   Subnet: 255.255.255.0

   Gateway: 192.168.10.1

   PC1:

   IP: 192.168.20.2

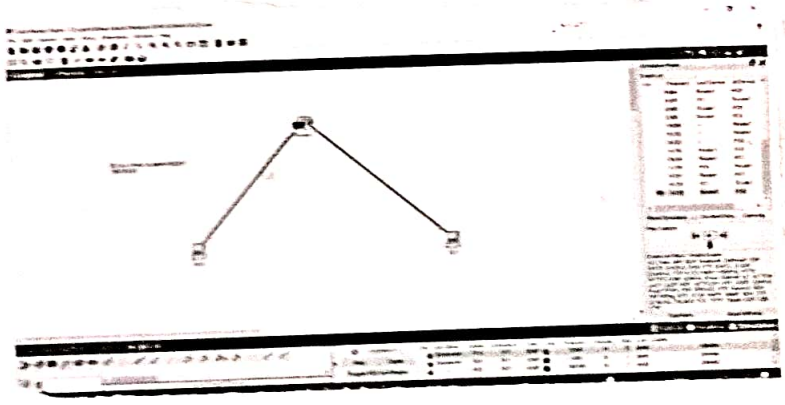
   Subnet: 255.255.255.0

   Gateway: 192.168.20.1

7. open command promt and type

   → ping: 192.168.20.2

8. Observe the packet and send PDU from PC0 to PC1 and observe packet transmission.



Result: Thus, internetworking using router is executed successfully.

Ex-11

Routing at network layer.

Aim:

To simulate static routing and RIP using cisco packet tracer

A) Static routing :-

Procedure

1. open cisco packet tracer
2. Place required routers, switch and PC
3. Connect the device using cable
4. Assign IP address to all PCs and routers
5. open the router CLI
6. open the following command to configure router
7. Use command to check routing table
   Show IP route.
8. Use ping command to check connectivity.

B) RIP routing :-

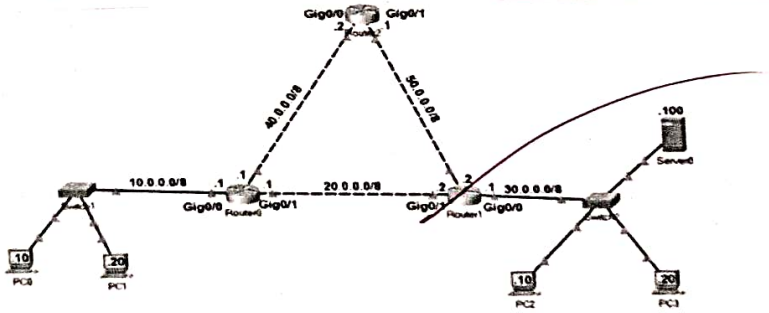
Procedure

1. Create network with 3 routers and PCs in packet tracer
2. Assign IP address to PCs and router interfaces
3. Configure the routes interface.
4. Enable RIP on router.

5. Repeat RIP configuration for other routers with their connected network.
6. Allow some time for routers to exchange info.
7. Check the routing table using show IP route.
8. Test communication using ping <destination IP>
9. Send simple PDU from source to destination PC and observe packet path.

Observation:

* In static, routers are manually configured by administration.
* In RIP, routers automatically exchange routing information.
* RIP selects a route based on low hop count.



Result:

Thus, static and RIP routing were successfully stimulated using cisco packet tracer.

Ex-11

End to End communication at Transport layer

Aim: To implement echo - client server using TCP and implement chat client - server using socket.

a) **Echo client - server :-**

```
import socket
import threading
def start_server():
    s = socket.socket (socket.AF_INET, socket.SOCK)
    s.bind ('127.0.0.1', 5000)
    s.listen (1)
    print ("listening server")
    conn, add = server.accept()
    print ("Connected by {add}")
    while True:
        d = conn.recv (1024).decode()
        if not data or data.lower() == 'quit':
            print ("{s} connection closed")
            break
        print ("Received: {data}")
        conn.send ("Server received : {data}")
conn.close()
server.close()

def start_client():
    c = socket.socket (socket.AF_INET,
                       socket.SOCK)
```

```
client.connect ('127.0.0.1", 5000)
print ("[client] connected)
while True:
    m = input ("enter msg:")
    client.send (m.encode())
    if m.lower() == 'quit':
        break
client.close().
```

Output:

```
(server) Listening on 127.0.0.5000 (client)
(server) Connected by ('127.0.0.1', 53748) enter msg:
hello
(Server) Received: hello
(Server response) Server received: hello
Enter msg: quit.
```

b) chat client - server :-

```
import socket
import threading
def udp_server():
    s = socket.socket (socket.AF_INET, socket.SOCK)
    s.bind ('127.0.0.1', 12000)
    print ("(server) running)
    while True:
        data, add = s.recform (1024)
        m = data.decode()
        if m.lower() == quit:
```

```
        print ("client 2addr 3 ended")
        break
    s.close()
    print ("stopped")
def udp_client():
    c = socket.socket (socket.AF_INET, socket.SOCK_DGRAM)
    s = ('127.0.0.1', 12000)
    while True:
        m = input ("enter msg:")
        c.sendto (m.encode(), server)
        if m.lower() == quit:
            print ("client) closing connection")
            break
        data, = c.recvfrom (1024)
    c.close()
print ("[client] stopped")
```

Output:

```
Enter msg : [Server] Running on 127.0.0.1.12000 Hi
[Server] received from ('127.0.0.1, 48670): hi
Echo from server: hi
Enter msg : hello
(Server) Received from ('127.0.0.1') : hello
Enter msg : quit
(client) closing connection
(client) stopped
```

Result: Thus end-end connection was successfully implemented.

Ex-13. Ping Program

Aim: To implement a ping program.

Code:

```
import os
import platform
import subprocess
def ping(host="google.com", count=4):
    param = "-n" if platform.system().lower()=="window"
                                        else "-c".
    comm = ["ping", param, str.(count), host]
    try:
        out = subprocess.check_output(command, uni_newline
                                        =True)
        print(out)
    except Exception as e:
        print("ping failed: {e}")
if __name__ == "__main__":
    ping("8.8.8.8")
```

Output:

```
ping 8.8.8.8 with 32 byte of data:
Reply from 8.8.8.8: bytes=32 time=6ms TTL=119
Reply from 8.8.8.8: bytes=32 time=9ms TTL=119
Ping stastistic for 8.8.8.8:
    Packets: Sent=4, Receive=4, Lost=0 (0% Loss),
Approx round trip times:
    Min = 5ms, Max = 9ms, Avg = 7ms.
```

Result: Thus ping program is implemented.

Ex-14 Raw Sockets Packet Sniffing

Aim:
To write a program using raw sockets to implement packet sniffing

Requirements:

→ Computer
→ python
→ Raw Socket support
→ Network interface
→ Administrator

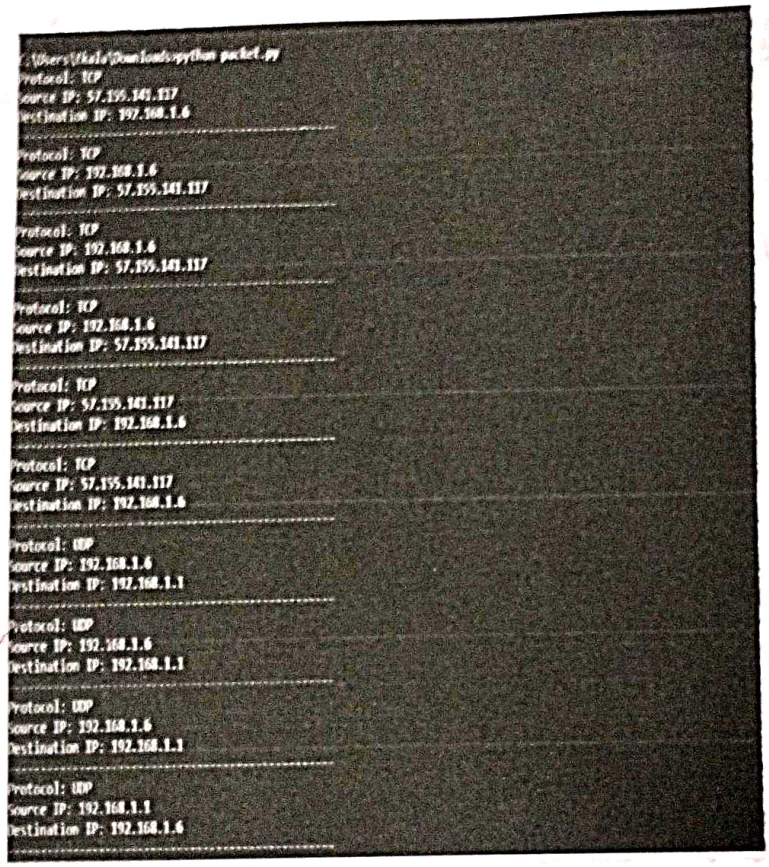
Algorithm:

1) Start program
2) Create raw sockets
3) Find the sockets
4) Wait for incoming network packets.
5) Capture packet
6) Extract packet info
7) Display packet detail

Procedure:

1) Open python IDE
2) Create python program
3) Import Socket module.
4) Create Raw socket

a) Use recvfrom() to receive packet
b) Read captured packet data
c) Display data



Result:

Thus Raw Socket based packet sniffing program was simplemented successfully.

Completed 13/10/25